

JS call(), bind(), apply() & Closures- Practice Questions

1. Using `call()` to Borrow a Method

Question:

Use the `call()` method to borrow a method from one object and use it in another.

Input:

```
const person1 = { name: "Alice" };  
const person2 = { name: "Bob" };
```

Expected Output:

```
Hello, my name is Alice  
Hello, my name is Bob
```

2. Using `apply()` to Pass an Array as Arguments

Question:

Use `apply()` to pass an array of arguments to a function. And add the two numbers and return the sum.

Expected Output:

```
15
```

3. Using `bind()` to Create a New Function

Question:

Use `bind()` to create a new function that has `this` permanently set to a specific object.

Input:

```
const user = { name: "Emma" };
```

Expected Output:

```
Hello, Emma
```

4. Using `call()` with Arguments

Question:

Use `call()` to pass arguments to a function. parameters should be name, age, country

Expected Output:

```
Liam is 25 years old from USA.
```

5. Using `apply()` to Merge Two Arrays

Question:

Use `apply()` to merge two arrays.

Input:

```
const arr1 = [1, 2, 3];  
const arr2 = [4, 5, 6];
```

Expected Output:

```
[1, 2, 3, 4, 5, 6]
```

6. Using `apply()` to Find Maximum in an Array

Question:

Use `apply()` to find the maximum number in an array.

Input:

```
const numbers = [3, 7, 2, 9, 5];
```

Expected Output:

```
9
```

7. Using `bind()` to Fix `this` Inside a Method

Question:

Fix the `this` reference inside an object method using `bind()`.

Input:

```
const car = {  
  brand: "Toyota",  
  getBrand: function () {  
    console.log(this.brand);  
  },  
};
```

Expected Output:

```
Toyota
```

8. Using Closures for Counter

Question:

Create a closure-based counter function.

Expected Output:

```
1  
2  
3
```

9. Using `bind()` for Delayed Execution

Question:

Use `bind()` to delay with 2 secs execution of a function.

Expected Output (after 2 seconds):

```
Hello, John!
```

10. Using `apply()` to multiply numbers

Question:

Write a function that takes an array of numbers and a context object. Use `apply()` to multiply all numbers and return the result.

Input:

```
const numbers = [2, 3, 4];
```

Expected Output:

```
24
```

JS `call()`, `bind()`, `apply()` & Closures- Assignment Questions

1. Calculate the area of a rectangle using `call()`

Write a function that calculates the area of a rectangle using `call()`.

Input:

```
length = 10, width = 5
```

Expected Output:

```
50
```

2. Using `bind()` to pre-set tax calculation

Write a function that pre-sets a tax percentage using `bind()`. For 1000/- i need calculate 15% tax

Expected Output:

```
150
```

3. Using `call()` to sum digits of a number

Write a function that sums the digits of a number using `call()`.

Input:

```
number = 1234
```

Expected Output:

```
10
```

4. Using `bind()` to create a power function

Write a function that binds a base number to an exponent function.

Input:

```
4
```

Expected Output:

```
16
```

12. Using closures to generate a discount calculator

Write a closure-based function that applies a 10% discount percentage to a given price is 500.

Expected Output:

```
450
```